SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4 **Total Marks: 20**

Code of Conduct

I S.Jenefar (192511345) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sf

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Q1)

CLOUD COMPUTING FUNDAMENTALS

CLOUD COMPUTING FUNDAMENTALS

DEFINITION

Provides on demand access to shared computing resources over the internet

CHARACTERISTICS

- On demand Self service
- Broad Network access
- Resource pooling
- Rapid elasticity

BENEFITS

- Cost efficiency
- Scalability
- flexibility
- Accessibility
- High availability

LIMITATION

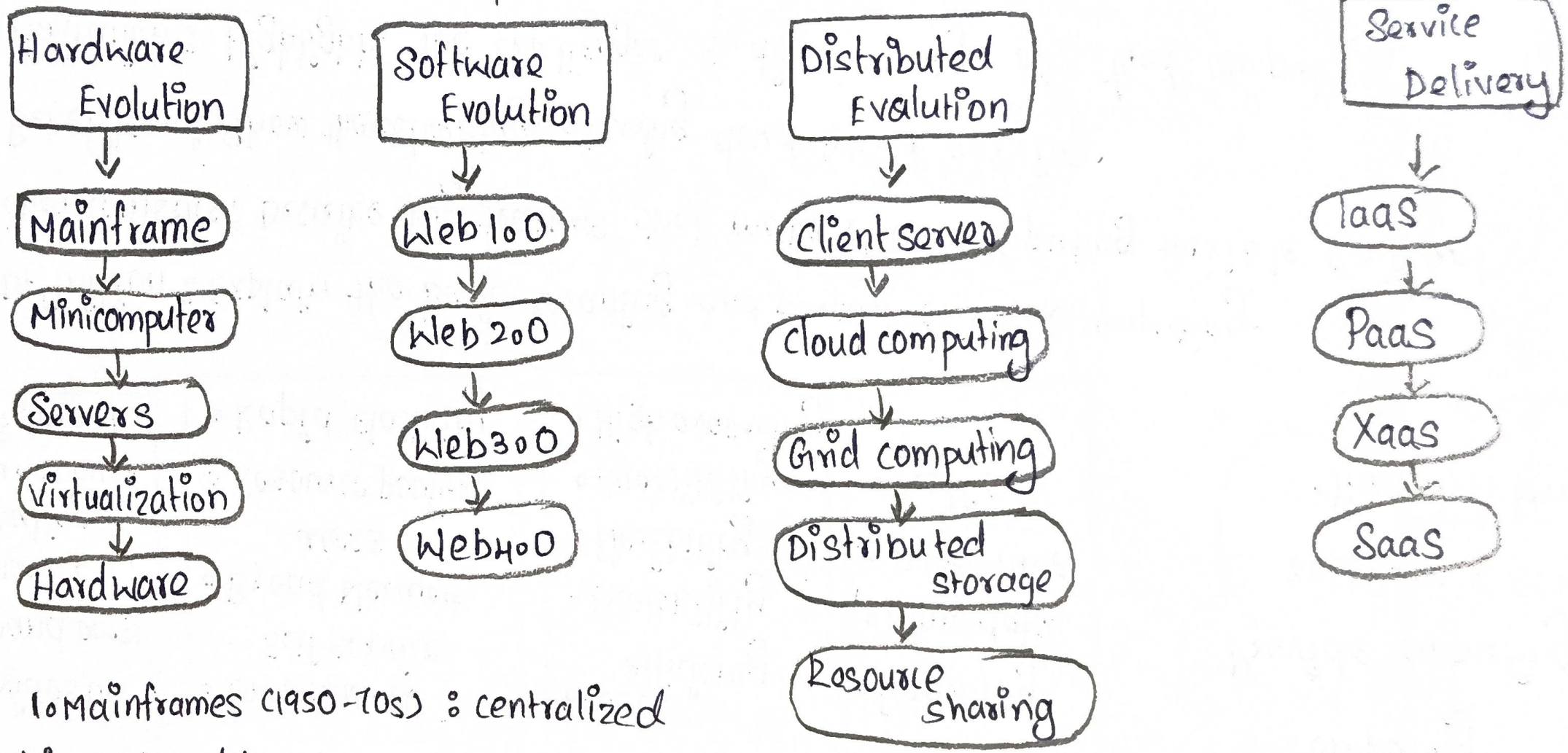
- Flexibility
- More data
- Access time
- Traffic

Cloud computing

Provide computing resources & servlie through internet

- Definition : Explain the basic meaning and purpose of cloud computing
- Characteristics : Describe key features that make cloud computing flexible & efficient
- Benefits : Show the advantage of using cloud based servlie
- Limitation : Highlights the challenges & disadvantage of Cloud computing

EVOLUTION OF CLOUD COMPUTING



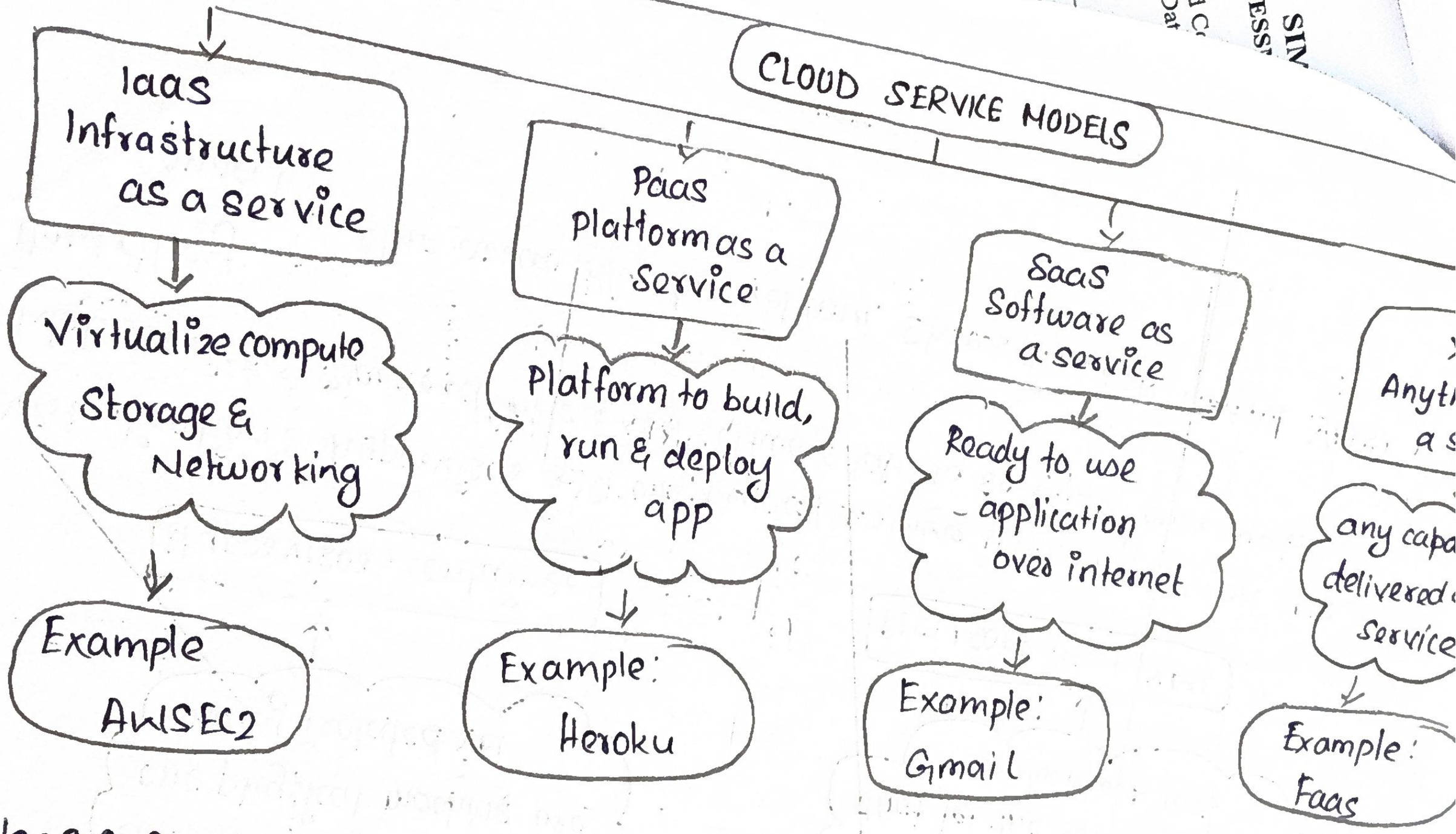
1. Mainframes (1950-70s) : centralized machines shared by many users

2. Client server & web (1980-90s): computing spread across networks

3. cluster and grid (1990s - 2000s) : Machines pooled their resources.

4. virtualization (2000s) : one machine ran many virtual machines

5. Cloud servile (2006 - present) : Iaas, Paas & Saas, Xaas.

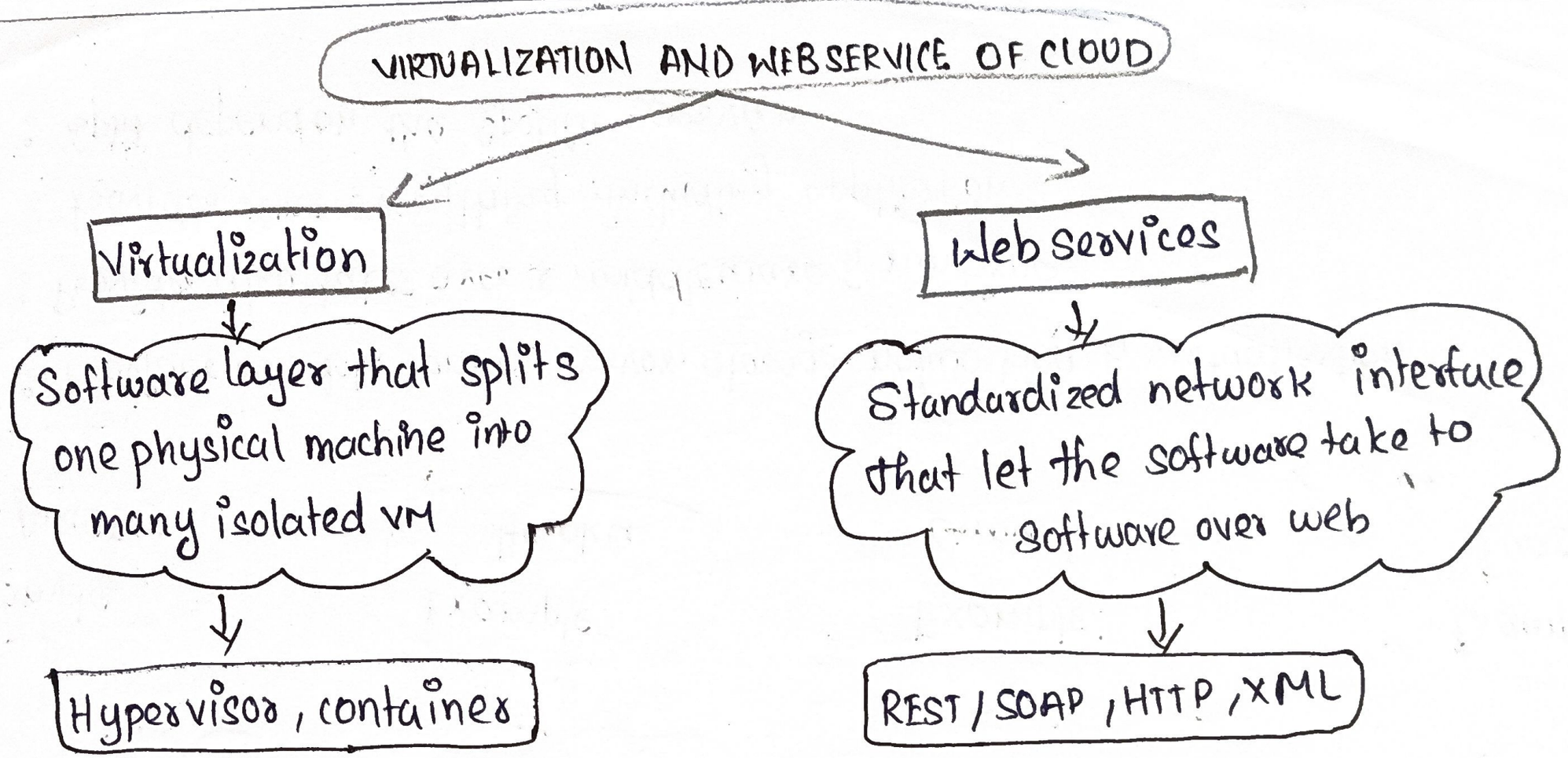


Iaas : Provider run data centers, server, storage, networking & virtualization

Paas : Provider also takes over os, middleware & run time

Saas : Provider runs everything, including application

Xaas : Split depend on the specific service.



- Virtualization: hypervisor split one physical machine into many isolated VMs.
- Webservice: Standard APIs & SOA software talk to software over web
- How they connect: The arrow b/w two column shows that APIs control VMs.